
ToE Letter IV — Mathematical Appendix

*Supplementary Derivations and Complete Mathematical Details for
Part I*

*Covering Sections Not Fully Addressed in the Main Text of Letter IV
Part I*

Author:	John Onimisi Obidi (jonimisiobidi@gmail.com)
Research Lab:	The Aether
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Preamble

Purpose of This Appendix. This document constitutes the Mathematical Appendix to *ToE Living Review Letter IV, Part I*. Its purpose is to furnish the complete mathematical details — including full proofs, coordinate derivations, and physical interpretations — for those sections of the Table of Contents of Letter IV Part I that are either

wholly absent from or only partially treated in the main text. The appendix is self-contained and may be read independently of the main letter, though it is intended as a companion document.

Scope. Letter IV Part I covers eight topic areas for which the main text either states results without proof, gives only special cases, or omits the topic entirely due to length constraints. This appendix supplies the missing material in full, section by section, preserving the original section numbering of the Letter IV Part I Table of Contents for cross-reference.

Structure. The appendix is organized into eight sections, labelled A through H, each supplementing a specific section of the main text:

- **Appendix A** (supplementing §2.6): *Why Mass, Pressure, Radiation, and Stress Are Not Alien to Information*. Full moment-theoretic derivation of all canonical macroscopic quantities from probability distributions.
- **Appendix B** (supplementing §3.3): *Smooth Maps, Diffeomorphisms, and Pullbacks — The Complete Theory*. Rigorous treatment of smooth maps, pushforwards, diffeomorphisms, pullback tensors, and Lie derivatives, with full relevance to the Obidi Metric construction.
- **Appendix C** (supplementing §4.1): *Scalars, Vectors, Covectors — A Unified Algebraic View*. Full algebraic classification of tensors via the tensor product, the universal property, musical isomorphisms, and the entropy field hierarchy.
- **Appendix D** (supplementing §4.9): *The Levi-Civita Connection — The Uniqueness Theorem*. Complete proof of the Fundamental

Theorem of Riemannian Geometry, including existence, uniqueness, the Koszul formula, and geodesic implications.

- **Appendix E** (supplementing §5.5): *How Fiber Integrals Produce Effective Fields — The Complete General Theory*. Full treatment of fiber integration on fiber bundles, the projection formula, Stokes' theorem for fiber integrals, coarse-graining, and the derivation of $T^{\mu\nu}$ from fiber integration.
- **Appendix F** (supplementing §6.5): *Higher Moments and Radiation — The Complete Theory*. Full moment hierarchy, heat flux, viscosity, radiation moments, Eddington factors, closure relations, and radiative transfer equations.
- **Appendix G** (supplementing §6.6): *Why Physical Tensors Are Moment Objects — The Systematic Theory*. The Tensor-Moment Correspondence Theorem, the metric as a zeroth moment, $T^{\mu\nu}$ as a second moment in Fourier space, and the Riemann tensor as a fourth moment.
- **Appendix H** (supplementing §6.7): *The Kinetic Theory Connection — Distribution Functions, the Boltzmann Equation, and the Stress-Energy Tensor*. Full relativistic Boltzmann equation, H-theorem, moment hierarchy, Chapman-Enskog expansion, and the derivation of the Einstein field equations as moment equations of the Entropic Boltzmann Equation.

Notation. Throughout, Greek indices ($\mu, \nu, \rho, \sigma, \lambda, \dots$) run over spacetime dimensions 0, 1, 2, 3. Latin indices ($i, j, k, \dots$) run over spatial dimensions 1, 2, 3. The metric signature is $(-, +, +, +)$. The Einstein summation convention is in force throughout: repeated upper-lower index pairs are summed. Natural units with $k_B = 1$ are used unless otherwise indicated.

The symbol ∇_μ denotes the covariant derivative compatible with the spacetime metric $g_{\mu\nu}$. The symbol ∂_μ denotes the ordinary partial derivative. Proofs are terminated with the symbol ■.

Appendix A — Supplementing §2.6

Why Mass, Pressure, Radiation, and Stress Are Not Alien to Information

This appendix provides a full, self-contained mathematical treatment demonstrating that mass, pressure, radiation, and stress — the four canonical macroscopic physical quantities — all emerge as moment structures of probability distributions, and are therefore irreducibly information-theoretic in character. This is not an analogy: it is a mathematical theorem.

A.1 — The Moment-Origin of Mass

Consider a localized matter distribution in three-dimensional space described by a mass density field $\rho(\mathbf{x})$, where $\rho(\mathbf{x}) \geq 0$ and $\rho(\mathbf{x}) \rightarrow 0$ rapidly as $|\mathbf{x}| \rightarrow \infty$. The total rest mass is the volume integral:

$$M = \int_V \rho(\mathbf{x}) d^3x \tag{A.1}$$

This is precisely the *zeroth-order moment* of the mass density distribution $\rho(\mathbf{x})$. No vector or tensor index appears — the result is a scalar (a (0,0)-tensor). In the language of moment theory, a zeroth moment is the total integral of the distribution, encoding its normalization.

In the Theory of Entropicity, the local mass density is identified with a component of the Entropic Stress-Energy Tensor $T^{\mu\nu}_{ent}$, which is itself obtained by the second-moment fiber integral over the statistical manifold (see Appendix E). Specifically:

$$\rho_{ent}(\mathbf{x}) = (1/c^2) T^{00}_{ent}(\mathbf{x}) \quad (\text{A.2})$$

where T^{00}_{ent} is the (0,0) component of the entropic stress-energy tensor, representing the local energy density divided by c^2 . The chain of derivation in ToE is:

ToE Mass Emergence Chain

Step 1: Binary entropy of a two-outcome experiment: $H_b(p) = -p \log_2 p - (1-p) \log_2(1-p)$

Step 2: Shannon information for a general distribution $\{p_i\}$: $H = -\sum_i p_i \log p_i$

Step 3: The Obidi Curvature Invariant (OCI) as the natural unit: $OCI = \ln(2)$ nats per binary digit

Step 4: The Obidi Action functional $A_{Obidi}[S]$ defined on the space of entropy fields

Step 5: The stationary solutions of A_{Obidi} are *entropic solitons* — localized, stable configurations of the entropic field

Step 6: The rest mass m of a particle is the zeroth moment (total integral) of the soliton's energy density: $m = (1/c^2) \int T^{00}_{ent} d^3x$

Thus mass is not a primitive ontological category. It is the zeroth moment of a localized probability distribution over entropic microstates. The localization

(soliton character) corresponds to the self-organizing property of the Obidi Action, and the total mass is simply the normalization of the localized energy distribution. This is equation (A.1) applied to the entropic context.

A.2 — The Moment-Origin of Pressure

Consider a gas of particles described by a kinetic distribution function $f(\mathbf{x}, \mathbf{p})$, where $f(\mathbf{x}, \mathbf{p}) d^3x d^3p$ is the expected number of particles in phase-space volume element $d^3x d^3p$ around position $\mathbf{x}$ and momentum $\mathbf{p}$. The isotropic (scalar) pressure is:

$$P = (1/3) \int (|\mathbf{p}|^2 / m) f(\mathbf{x}, \mathbf{p}) d^3p \div \int f(\mathbf{x}, \mathbf{p}) d^3p \quad (\text{A.3})$$

This is the trace of the second spatial moment of the momentum distribution. More precisely, the full pressure tensor (valid for relativistic particles) is:

$$P^{ij} = \int (\dot{p}^i \dot{p}^j) / (\gamma m) \cdot f(\mathbf{x}, \mathbf{p}) \cdot (d^3p / p^0) \quad (\text{A.4})$$

where $\gamma = p^0/mc$ is the Lorentz factor, $p^0 = \sqrt{(m^2c^4 + |\mathbf{p}|^2c^2)}/c$ is the relativistic energy divided by c , and the measure d^3p / p^0 is the Lorentz-invariant momentum-space measure (see §H.1). The scalar pressure is the trace:

$$P = (1/3) g_{ij} P^{ij} = (1/3) \delta_{ij} P^{ij} = (1/3)(P^{11} + P^{22} + P^{33}) \quad (\text{A.5})$$

In the non-relativistic limit, $\gamma \rightarrow 1$, $p^0 \rightarrow mc$, and equation (A.4) reduces to the classical kinetic pressure formula. The key observation is that P^{ij} is the *second moment* of the momentum distribution — it is a quadratic average of $\dot{p}^i \dot{p}^j$ weighted by f . Pressure is therefore not a primitive concept: it is a statement about the informational spread of the momentum distribution. A distribution

with large spread (high uncertainty in momentum) corresponds to high pressure; a distribution concentrated near zero momentum has low pressure.

A.3 — The Moment-Origin of Stress

The full stress tensor σ^{ij} is the *traceless* part of the second moment tensor P^{ij} :

$$\sigma^{ij} = P^{ij} - (1/3) \delta^{ij} P^{kk} \quad (\text{A.6})$$

where $P^{kk} = P^{11} + P^{22} + P^{33} = 3P$ is the trace. By construction, $\delta_{ij} \sigma^{ij} = 0$ — the stress tensor is traceless. The decomposition (A.6) splits the second moment into two irreducible parts:

- **Isotropic part:** $(1/3) \delta^{ij} P^{kk}$ — this is the pressure, equal in all spatial directions.
- **Anisotropic (traceless) part:** σ^{ij} — this encodes shear forces.

Physically: if the momentum distribution $f(\mathbf{x}, \mathbf{p})$ is isotropic (spherically symmetric in momentum space), then $P^{ij} = P \delta^{ij}$ and consequently $\sigma^{ij} = 0$. There is pressure but no shear stress. If f is anisotropic — if it has a preferred direction in momentum space — then $P^{ij} \neq P \delta^{ij}$ and $\sigma^{ij} \neq 0$. Shear stress is therefore purely a statement about the *shape* (the anisotropy) of the momentum distribution. This is a purely information-theoretic statement: it depends only on which directions are more probable in the momentum distribution, not on any primitive mechanical property of the medium.

A.4 — The Moment-Origin of Radiation

Radiation (electromagnetic or gravitational) is described by its stress-energy tensor. Let $I(\mathbf{n})$ be the *specific intensity* — the energy flux per unit area per unit time per unit solid angle per unit frequency interval, integrated over all frequencies — in the spatial direction specified by the unit three-vector $\mathbf{n}$. Define the null four-vector $n^\mu = (1, \mathbf{n})$. The radiation stress-energy tensor is:

$$T^{\mu\nu}_{rad} = (1/c) \int_{S^2} n^\mu n^\nu I(\mathbf{n}) d\Omega \quad (\text{A.7})$$

where the integral is over all directions on the unit sphere S^2 . This is manifestly the second moment of the directional distribution $I(\mathbf{n})$. The radiation pressure tensor is the spatial part:

$$P^j_{rad} = (1/c) \int_{S^2} n^j n^j I(\mathbf{n}) d\Omega \quad (\text{A.8})$$

and the radiation energy density is:

$$u_{rad} = T^{00}_{rad} = (1/c) \int_{S^2} I(\mathbf{n}) d\Omega \quad (\text{A.9})$$

The *Eddington factor tensor* is the normalized second moment:

$$f^{ij} = P^j_{rad} / u_{rad} \quad (\text{A.10})$$

The Eddington factor encodes the angular distribution of radiation. Its eigenvalues lie in $[0, 1]$. Two limiting cases:

- **Isotropic radiation** (thermal equilibrium): $I(\mathbf{n}) = I_0$ (constant), so $\int n^i n^j d\Omega = (4\pi/3) \delta^{ij}$, giving $f^{ij} = (1/3) \delta^{ij}$. This is the *Eddington limit*.
- **Unidirectional beam** (free streaming): $I(\mathbf{n}) = I_0 \delta^2(\mathbf{n} - \mathbf{n}_0)$, giving $f^{ij} = n^i_0 n^j_0$. Maximum anisotropy.

The transition from isotropic to anisotropic Eddington factor is a purely informational statement: it describes the shape of the probability distribution of photon propagation directions. The amount of radiation pressure, its directionality, and its shear stress content are all encoded in the moments of $I(\mathbf{n})$. Radiation, too, is information.

A.5 — The Unified Information-Theoretic Origin

Fundamental Principle (Moment-Origin of Physical Quantities)

Every macroscopic physical quantity appearing in the stress-energy tensor $T^{\mu\nu}$ is a moment of an underlying probability distribution. Since moments are purely information-theoretic constructs — they are defined by integrals of powers of a random variable against a distribution — all macroscopic physics is information-theoretic in origin. This is not a metaphor. It is a mathematical theorem.

The following table provides the complete classification:

Physical Quantity	Moment Order	Distribution	Index Structure
Rest Mass M	0th (zeroth)	Mass/energy density distribution	Scalar — (0,0) tensor
Momentum / Current N^μ	1st (first)	Momentum distribution	Vector — (1,0) tensor
Pressure P	2nd (trace)	Momentum distribution	Scalar (trace of symmetric 2-tensor)
Stress σ^{ij}	2nd (traceless)	Momentum distribution	Symmetric traceless (0,2) tensor
Radiation $T^{\mu\nu}_{rad}$	2nd (second)	Directional intensity distribution	Symmetric (2,0) tensor
Viscosity η, ζ	2nd (off-diagonal)	Velocity gradient distribution	Scalar coupling constants of tensor

In ToE, the entropic stress-energy tensor $T^{\mu\nu}_{ent}$ is the second moment of the entropic distribution function $f_{ent}(\mathbf{x}, q)$ over the entropic momentum manifold (the cotangent bundle of the statistical manifold Λ). The full content of §2.6 of Letter IV Part I — that mass, pressure, radiation, and stress are not alien to information — is the claim that all these quantities are moment objects of the underlying entropic distribution. Appendix G provides the systematic theory; Appendix H derives these moments explicitly from the Entropic Boltzmann Equation.

Appendix B — Supplementing §3.3

Smooth Maps, Diffeomorphisms, and Pullbacks — The Complete Theory

B.1 — Smooth Maps Between Manifolds

Let M be a smooth manifold of dimension m and N be a smooth manifold of dimension n . A *smooth map* is a function $\varphi: M \rightarrow N$ such that for every point $p \in M$, there exist coordinate charts (U, x) around p in M and (V, y) around $\varphi(p)$ in N with $\varphi(U) \subseteq V$, and the coordinate representation:

$$y^a = \varphi^a(x^1, \dots, x^m), \quad a = 1, \dots, n$$

(B.1)

consists of n smooth (infinitely differentiable) functions of the m coordinates on M . The *Jacobian matrix* of φ at a point p is the $n \times m$ matrix of partial derivatives:

$$J^a_{\mu}(p) = \partial \varphi^a / \partial x^{\mu} |_p, \quad a = 1, \dots, n; \mu = 1, \dots, m$$

(B.2)

The *rank* of φ at p is the rank of $J_\mu(p)$ as a matrix. A point $p \in M$ is:

- A *regular point* if $\text{rank}(J) = \min(m, n)$ at p ;
- A *critical point* if $\text{rank}(J) < \min(m, n)$ at p .

The image of a critical point under φ is called a *critical value*. Sard's theorem guarantees that the set of critical values has measure zero in N — "most" points of N are regular values.

B.2 — The Differential (Pushforward)

The *differential* (or *pushforward*) of φ at a point $p \in M$ is the linear map:

$$d\varphi_p : T_p M \rightarrow T_{\varphi(p)} N$$

(B.3)

from the tangent space at p to the tangent space at $\varphi(p)$. For a tangent vector $X_p = X^\mu (\partial/\partial x^\mu)|_p \in T_p M$, the pushforward is defined by the Jacobian action:

$$(d\varphi_p(X))^\alpha = J^\alpha_\mu X^\mu = (\partial\varphi^\alpha / \partial x^\mu) X^\mu$$

(B.4)

This is the coordinate-free formulation of the chain rule of multivariable calculus. For a smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ with $\gamma(0) = p$ and tangent vector $\gamma'(0) = X_p$, the pushforward is the tangent vector to the image curve $\varphi \circ \gamma$ at $t = 0$:

$$(d\varphi_p(X))^\alpha = d/dt [\varphi^\alpha(\gamma(t))]_{t=0}$$

(B.5)

Equations (B.4) and (B.5) are equivalent: (B.5) is the intrinsic (coordinate-free) definition; (B.4) is its coordinate expression. The pushforward is covariant: it goes from M to N in the same direction as φ .

B.3 — Immersions, Submersions, and Embeddings

Three important classes of smooth maps are defined by the rank condition on the differential:

- **Immersion:** $d\varphi_p$ is injective (one-to-one) at every $p \in M$. This requires $\text{rank}(J) = m$, hence $m \leq n$. Locally, an immersion looks like the inclusion of $\mathbb{R}^m$ into $\mathbb{R}^n$. Every direction in M maps to a distinct direction in N — no direction is "collapsed."
- **Submersion:** $d\varphi_p$ is surjective (onto) at every $p \in M$. This requires $\text{rank}(J) = n$, hence $n \leq m$. Locally, a submersion looks like the projection of $\mathbb{R}^m$ onto $\mathbb{R}^n$.
- **Embedding:** An immersion $\varphi: M \rightarrow N$ that is additionally a homeomorphism onto its image $\varphi(M)$ (with the subspace topology from N). An embedding places a copy of M inside N without self-intersection and without limit points that are not in the image.

Theorem B.1 (Nash Embedding Theorem, Nash 1956)

Every smooth Riemannian manifold (M, g) can be isometrically embedded into Euclidean space $\mathbb{R}^N$ for some sufficiently large integer N . That is, there exists a smooth embedding $\varphi: M \rightarrow \mathbb{R}^N$ such that the pullback metric satisfies $\varphi^* \tilde{g}_{Eucl} = g$, where $\tilde{g}_{Eucl}$ is the standard Euclidean metric on $\mathbb{R}^N$.

Relevance to ToE: The statistical manifold Λ of entropic microstates is a Riemannian manifold equipped with the Fisher-Rao metric G^F_{AB} . By the Nash Embedding Theorem, Λ embeds isometrically into an ambient Euclidean space $\mathbb{R}^N$. The Obidi Metric $\tilde{g}_{\mu\nu}$ on the physical spacetime manifold M is the pullback of the ambient Euclidean metric via the emergence map $\varphi: \Lambda \rightarrow \mathbb{R}^N$, modified by the Obidi Transformation (see §B.5). This is why the spacetime

metric is said to be induced by the information geometry of the statistical manifold.

B.4 — Diffeomorphisms

A *diffeomorphism* is a smooth bijection $\varphi: M \rightarrow N$ whose inverse $\varphi^{-1}: N \rightarrow M$ is also smooth. If a diffeomorphism exists between M and N , they are called *diffeomorphic*: they are the same smooth manifold up to a smooth relabelling of points. The set of all diffeomorphisms from M to itself forms a group under composition, denoted $Diff(M)$, which is an infinite-dimensional Lie group.

Characterization of Diffeomorphisms

A smooth map $\varphi: M \rightarrow N$ is a diffeomorphism if and only if the differential $d\varphi_p: T_p M \rightarrow T_{\varphi(p)} N$ is a linear isomorphism at every point $p \in M$. Equivalently: $m = n$ (equal dimensions) and $\det(J^\varphi_\mu) \neq 0$ everywhere on M .

Diffeomorphism Invariance in General Relativity and ToE. The invariance of a physical theory under the group $Diff(M)$ — called *diffeomorphism invariance* or *general covariance* — is the statement that the physics does not depend on the choice of coordinate chart. Formally: for any diffeomorphism $\varphi \in Diff(M)$, the pullback φ^*g of the metric satisfies the same field equations as g . This is the precise mathematical content of Einstein's principle of general covariance.

In ToE, diffeomorphism invariance of the Obidi Action A_{Obidi} under $Diff(\Lambda)$ (diffeomorphisms of the statistical manifold) is the statement that the entropic physics is coordinate-independent on the statistical manifold. Via the Entropic Noether Principle (ENP), this invariance generates conserved entropic currents — see §B.6.

B.5 — The Pullback of Tensors

Let $\varphi: M \rightarrow N$ be a smooth map and let T be a covariant tensor field of type $(0, s)$ on N . The *pullback* φ^*T is a covariant tensor field of type $(0, s)$ on M , defined by:

$$(\varphi^*T)_p(X_1, \dots, X_s) = T_{\varphi(p)}(d\varphi_p(X_1), \dots, d\varphi_p(X_s)) \quad (\text{B.6})$$

for tangent vectors $X_1, \dots, X_s \in T_pM$. In coordinates: if $T = T_{a_1 \dots a_s}(y) dy^{a_1} \otimes \dots \otimes dy^{a_s}$ on N , then:

$$(\varphi^*T)_{\mu_1 \dots \mu_s}(X) = T_{a_1 \dots a_s}(\varphi(X)) \cdot (\partial\varphi^{a_1}/\partial x^{\mu_1}) \cdot \dots \cdot (\partial\varphi^{a_s}/\partial x^{\mu_s}) \quad (\text{B.7})$$

This is the component formula for the pullback. Note the key feature: the pullback goes *backwards* (from N to M), opposite to the direction of φ . This is expressed by saying the pullback is *contravariantly natural*.

The **pullback of the metric** is of particular importance. If (N, h) is a Riemannian manifold and $\varphi: M \rightarrow N$ is a smooth map, the pullback metric on M is:

$$(\varphi^*h)_{\mu\nu}(x) = h_{ab}(\varphi(x)) \cdot (\partial\varphi^a/\partial x^\mu) \cdot (\partial\varphi^b/\partial x^\nu) \quad (\text{B.8})$$

The pullback metric (B.8) is positive definite (Riemannian) if and only if φ is an immersion — so that $d\varphi_p$ is injective and no direction in T_pM is collapsed to zero in $T_{\varphi(p)}N$.

ToE Application: The Obidi Metric as a Pullback

The Obidi Metric $\tilde{g}_{\mu\nu}$ on the information manifold Λ is the pullback of the ambient Euclidean metric via the emergence map $\varphi: \Lambda \rightarrow \mathbb{R}^N$, subsequently

modified by the Obidi Transformation $\mathcal{T}_{Obidi}$. Equation (B.8) is the explicit mathematical formula implementing the statement that "the spacetime metric is induced by the information geometry of the statistical manifold."

B.6 — The Lie Derivative

Let X be a smooth vector field on M and let $\Phi_t: M \rightarrow M$ be its flow (the one-parameter family of diffeomorphisms generated by X). The *Lie derivative* of a tensor field T along X is:

$$\mathcal{L}_X T = \lim_{t \rightarrow 0} [(\Phi_t)^* T - T] / t \quad (\text{B.9})$$

For a covariant (0,2) tensor field $T_{\mu\nu}$, the coordinate expression of the Lie derivative is:

$$(\mathcal{L}_X T)_{\mu\nu} = X^\rho \partial_\rho T_{\mu\nu} + T_{\rho\nu} \partial_\mu X^\rho + T_{\mu\rho} \partial_\nu X^\rho \quad (\text{B.10})$$

For the metric tensor $g_{\mu\nu}$, using the Levi-Civita connection (see Appendix D), the Lie derivative takes the particularly important form:

$$(\mathcal{L}_X g)_{\mu\nu} = \nabla_\mu X_\nu + \nabla_\nu X_\mu = 2\nabla_{(\mu} X_{\nu)} \quad (\text{B.11})$$

A vector field X is a *Killing vector field* if $\mathcal{L}_X g = 0$, i.e., if:

$$\nabla_{(\mu} X_{\nu)} = 0 \quad (\text{Killing's equation}) \quad (\text{B.12})$$

Killing vector fields generate *isometries* — diffeomorphisms that preserve the metric. By Noether's theorem, each Killing vector field corresponds to a conserved current. The classical conservation laws of energy (time translation invariance: $X = \partial/\partial t$), linear momentum (spatial translation invariance: $X = \partial/\partial x^i$), and angular momentum (rotational invariance) all correspond to Killing vectors of the Minkowski metric.

In ToE: The Entropic Noether Principle (ENP) states that every symmetry of the Obidi Action — formulated as a diffeomorphism of the statistical manifold leaving the action invariant — corresponds to a conserved entropic current. The Lie derivative (B.11) is the precise mathematical tool quantifying how the Obidi Metric changes under the flow of a vector field on Λ . When this change vanishes (Killing condition (B.12)), the symmetry is exact and a conservation law holds.

Appendix C — Supplementing §4.1

Scalars, Vectors, Covectors — A Unified Algebraic View

C.1 — The Algebra of Tensor Products

At each point p of a smooth manifold M ($\dim n$), two canonical vector spaces are defined: the *tangent space* $T_p M$ (spanned by the basis vectors $\partial/\partial x^\mu|_p$) and the *cotangent space* $T_p^* M$ (spanned by the dual basis $dx^\mu|_p$). These spaces form a *dual pair* under the natural bilinear pairing:

$$\langle \omega, X \rangle = \omega(X) \in \mathbb{R}, \quad \omega \in T_p^* M, \quad X \in T_p M \quad (\text{C.1})$$

This pairing is bilinear and non-degenerate. Algebraically, $T_p^* M = (T_p M)^*$, the dual vector space. In coordinates: $\langle dx^\mu, \partial/\partial x^\nu \rangle = \delta^\mu_\nu$.

The full algebra of tensors at p is constructed by taking tensor products. The space of type (r, s) tensors at p is:

$$T_s(p) = T_p M \otimes \dots \otimes T_p M (r \text{ copies}) \otimes T_p^* M \otimes \dots \otimes T_p^* M (s \text{ copies}) \quad (\text{C.2})$$

This space has dimension n^{r+s} and possesses a basis $\{\partial/\partial x^{\mu_1} \otimes \dots \otimes \partial/\partial x^{\mu_r} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_s}\}$. A general type (r, s) tensor has components $T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s}$ in this basis.

C.2 — The Universal Property of the Tensor Product

The tensor product $T_p M \otimes T_p^* M$ is characterized (up to canonical isomorphism) by the following universal property:

Universal Property of the Tensor Product

For any bilinear map $B: T_p M \times T_p^* M \rightarrow V$ to any vector space V , there exists a unique linear map $\tilde{B}: T_p M \otimes T_p^* M \rightarrow V$ such that $B = \tilde{B} \circ \otimes$, where $\otimes: T_p M \times T_p^* M \rightarrow T_p M \otimes T_p^* M$ is the canonical bilinear map sending (X, ω) to $X \otimes \omega$.

This universal property makes the tensor product *canonical*: it is the most efficient way to linearize bilinear maps. The physical significance is that all bilinear pairings in physics (inner products, contractions, couplings) factor canonically through tensor products — this is why tensors are the natural language of physical law.

C.3 — Unified Classification Table

Type	Name	Index Slots	Dimension	Example in Physics
$(0, 0)$	Scalar	None	1	Temperature T , Entropy S ,

Type	Name	Index Slots	Dimension	Example in Physics
				OCI = $\ln(2)$
$(1, 0)$	Vector / Contravariant vector	1 upper index V^μ	n	Four-velocity u^μ , current J^μ
$(0, 1)$	Covector / One-form / Covariant vector	1 lower index ω_μ	n	Entropy gradient $\partial_\mu S$, potential A_μ
$(2, 0)$	Contravariant 2-tensor	2 upper indices $T^{\mu\nu}$	n^2	Inverse metric $g^{\mu\nu}$
$(0, 2)$	Covariant 2-tensor	2 lower indices $T_{\mu\nu}$	n^2	Metric $g_{\mu\nu}$, Faraday tensor $F_{\mu\nu}$
$(1, 1)$	Mixed 2-tensor	1 up, 1 down: $T^\mu{}_\nu$	n^2	Kronecker delta $\delta^\mu{}_\nu$, stress tensor $T^\mu{}_\nu$
$(0, 3)$	Covariant 3-tensor	3 lower indices T_{ijk}	n^3	Amari-Čencov cubic tensor C_{ijk}
$(1, 3)$	Mixed 4-tensor	1 up, 3 down: $T^\rho{}_{\sigma\mu\nu}$	n^4	Riemann tensor $R^\rho{}_{\sigma\mu\nu}$

C.4 — The Canonical Identification: Musical Isomorphisms

On a smooth manifold without a metric, $T_p M$ and $T_p^* M$ are distinct vector spaces of the same dimension — they are abstractly isomorphic, but *not canonically so* (any isomorphism requires a choice). A metric $g_{\mu\nu}$ provides a *canonical* isomorphism, called the *musical isomorphisms*:

The **flat** map ($\flat$, lowering an index):

$$\flat : T_p M \rightarrow T_p^* M, X^\mu \mapsto X_\nu = g_{\mu\nu} X^\mu$$

(C.3)

The **sharp** map ($\sharp$, raising an index):

$$\sharp: T_p^*M \rightarrow T_pM, \omega_\mu \mapsto \omega^\nu = g^{\mu\nu} \omega_\mu \quad (\text{C.4})$$

These isomorphisms are inverses: $\sharp \circ \flat = id_{T_pM}$ and $\flat \circ \sharp = id_{T_p^*M}$. They depend explicitly on the metric: change the metric and the identification changes. In the absence of a metric, there is *no* canonical identification of vectors and covectors.

This is why the entropy gradient $\partial_\mu S$ is naturally a *covector* (a one-form, living in T_p^*M): it is defined by the action of S on tangent vectors via the directional derivative, which requires no metric. By contrast, the entropic current $J^\mu_{ent} = g^{\mu\nu} \partial_\nu S$ is a *vector* (in T_pM): its definition requires raising the index via the metric, which requires the Obidi Metric to already be in place.

C.5 — The Contraction Operation and Index Saturation

A *contraction* pairs one upper and one lower index of a tensor, saturating them by summation. For a type (r, s) tensor $T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s}$, contracting the k -th upper index with the l -th lower index gives a type $(r-1, s-1)$ tensor:

$$C^{\mu_1 \dots \hat{\mu}_k \dots \mu_r}_{\nu_1 \dots \hat{\nu}_l \dots \nu_s} = \sum_{\rho=0}^{n-1} T^{\mu_1 \dots \rho \dots \mu_r}_{\nu_1 \dots \rho \dots \nu_s} \quad (\text{C.5})$$

Important special cases:

- Trace of the metric: $g^{\mu\nu} g_{\mu\nu} = \delta^\mu_\mu = n$ (the spacetime dimension).
- Ricci tensor: $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$ (contraction of Riemann on first and third indices).
- Scalar curvature: $R = g^{\mu\nu} R_{\mu\nu}$ (double contraction).

C.6 — Why the Algebraic View Matters for ToE

The unified algebraic view reveals why the entropy field S admits three distinct algebraic levels, with different metric requirements:

- **Level 0:** $S \in T^0_0$ — a scalar. No metric is needed. S is defined as the information content of the probability distribution, independent of any spacetime structure.
- **Level 1:** $\partial_\mu S \in T^0_1$ — a covector. No metric is needed. The exterior derivative $dS = \partial_\mu S \lrcorner dx^\mu$ is defined purely by the differentiable structure of the manifold.
- **Level 2:** $\nabla^\mu S = g^{\mu\nu} \partial_\nu S \in T^1_0$ — a vector. The metric is required to raise the index. This step invokes the sharp isomorphism (C.4).

The hierarchy Level 0 $\rightarrow$ Level 1 is metric-independent (it uses only the exterior derivative). The step Level 1 $\rightarrow$ Level 2 requires the Obidi Metric (via the sharp isomorphism). This is why, in the ToE construction, the Obidi Transformation — which establishes $\tilde{g}_{\mu\nu}$ from the Fisher-Rao metric — must logically precede any computation involving the entropic current as a vector field. The order of mathematical constructions in ToE is not arbitrary: it is dictated by the algebraic structure of tensor spaces.

Appendix D — Supplementing §4.9

The Levi-Civita Connection — The Uniqueness Theorem (Fundamental Theorem of Riemannian Geometry)

D.1 — Statement of the Theorem

Theorem D.1 (Fundamental Theorem of Riemannian Geometry)

On any pseudo-Riemannian manifold (M, g) , there exists a unique affine connection ∇ satisfying simultaneously:

(i) Torsion-free (symmetric): $\nabla_X Y - \nabla_Y X = [X, Y]$ for all smooth vector fields X, Y on M .

(ii) Metric compatible: $\nabla_Z g = 0$ for all smooth vector fields Z , equivalently: $Z(g(X, Y)) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$ for all X, Y, Z .

This unique connection is the *Levi-Civita connection*, with Christoffel symbols:

$$\Gamma_{\mu\nu}^{\rho} = (1/2) g^{\rho\lambda} (\partial_{\mu} g_{\nu\lambda} + \partial_{\nu} g_{\mu\lambda} - \partial_{\lambda} g_{\mu\nu})$$

D.2 — Proof of Existence

Proof (Existence).

We verify that the Christoffel symbols $\Gamma_{\mu\nu}^{\rho} = (1/2) g^{\rho\lambda} (\partial_{\mu} g_{\nu\lambda} + \partial_{\nu} g_{\mu\lambda} - \partial_{\lambda} g_{\mu\nu})$ define a connection satisfying (i) and (ii).

Step 1 — Torsion-free. The torsion tensor is:

$$T_{\mu\nu}^{\rho} = \Gamma_{\mu\nu}^{\rho} - \Gamma_{\nu\mu}^{\rho}$$

(D.1)

Since the Christoffel symbols are symmetric in the lower indices — evident from the formula, since $\partial_{\mu} g_{\nu\lambda} + \partial_{\nu} g_{\mu\lambda} - \partial_{\lambda} g_{\mu\nu}$ maps to $\partial_{\nu} g_{\mu\lambda} + \partial_{\mu} g_{\nu\lambda} - \partial_{\lambda} g_{\nu\mu}$ under interchange $\mu \leftrightarrow \nu$, and since $g_{\mu\nu} = g_{\nu\mu}$ — we have $\Gamma_{\mu\nu}^{\rho} = \Gamma_{\nu\mu}^{\rho}$. Therefore $T_{\mu\nu}^{\rho} = 0$. ✓

Step 2 — Metric compatibility. We must verify $\nabla_{\rho} g_{\mu\nu} = 0$. Compute:

$$\nabla_\rho g_{\mu\nu} = \partial_\rho g_{\mu\nu} - \Gamma^\lambda_{\rho\mu} g_{\lambda\nu} - \Gamma^\lambda_{\rho\nu} g_{\mu\lambda}$$

(D.2)

Substituting the Christoffel formula into $\Gamma^\lambda_{\rho\mu} g_{\lambda\nu} = (1/2)(\partial_\rho g_{\mu\nu} + \partial_\mu g_{\rho\nu} - \partial_\nu g_{\rho\mu})$ and $\Gamma^\lambda_{\rho\nu} g_{\mu\lambda} = (1/2)(\partial_\rho g_{\nu\mu} + \partial_\nu g_{\rho\mu} - \partial_\mu g_{\rho\nu})$:

$$\nabla_\rho g_{\mu\nu} = \partial_\rho g_{\mu\nu} - (1/2)(\partial_\rho g_{\mu\nu} + \partial_\mu g_{\rho\nu} - \partial_\nu g_{\rho\mu}) - (1/2)(\partial_\rho g_{\nu\mu} + \partial_\nu g_{\rho\mu} - \partial_\mu g_{\rho\nu})$$

(D.3)

Expanding and using $g_{\mu\nu} = g_{\nu\mu}$:

$$= \partial_\rho g_{\mu\nu} - (1/2)\partial_\rho g_{\mu\nu} - (1/2)\partial_\mu g_{\rho\nu} + (1/2)\partial_\nu g_{\rho\mu} - (1/2)\partial_\rho g_{\mu\nu} - (1/2)\partial_\nu g_{\rho\mu} + (1/2)\partial_\mu g_{\rho\nu}$$

(D.4)

$$= \partial_\rho g_{\mu\nu} - \partial_\rho g_{\mu\nu} = 0$$

(D.5)

Metric compatibility is satisfied. ✓



D.3 — Proof of Uniqueness

Proof (Uniqueness).

Suppose ∇ and ∇' are two affine connections on (M, g) both satisfying (i) and (ii). Define their difference tensor:

$$D(X, Y) = \nabla_X Y - \nabla'_X Y$$

(D.6)

The difference of two connections is always a (1,2) tensor field (the connection-transformation law guarantees that inhomogeneous terms cancel). We derive two properties of D :

Property 1 (from metric compatibility): For any vector fields X, Y, Z :

$$Zg(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y) = g(\nabla'_Z X, Y) + g(X, \nabla'_Z Y)$$

(D.7)

Subtracting: $0 = g(D(Z, X), Y) + g(X, D(Z, Y))$, giving:

$$g(D(Z, X), Y) = -g(X, D(Z, Y)) \dots (\star)$$

(D.8)

Property 2 (from torsion-freeness):

$$D(X, Y) - D(Y, X) = (\nabla_X Y - \nabla'_X Y) - (\nabla_Y X - \nabla'_Y X) = [X, Y] - [X, Y] = 0$$

(D.9)

So $D(X, Y) = D(Y, X)$ — the difference tensor is symmetric. ...(\star\star)

Conclusion. Using (\star) and (\star\star) cyclically:

$$g(D(X, Y), Z) = -g(Y, D(X, Z)) \text{ [by } (\star), Z \rightarrow X, X \rightarrow Y, Y \rightarrow Z]$$

(D.10)

$$= -g(Y, D(Z, X)) \text{ [by } (\star\star)]$$

(D.11)

$$= +g(D(Y, Z), X) \text{ [by } (\star)]$$

(D.12)

$$= +g(D(Z, Y), X) \text{ [by } (\star\star)]$$

(D.13)

$$= -g(Z, D(Y, X)) \text{ [by } (\star)]$$

(D.14)

$$= -g(Z, D(X, Y)) \text{ [by } (\star\star)]$$

(D.15)

Therefore: $g(D(X, Y), Z) = -g(D(X, Y), Z)$, which implies $g(D(X, Y), Z) = 0$ for all Z . Since g is non-degenerate, $D(X, Y) = 0$ for all X, Y . Thus $\nabla = \nabla'$.



D.4 — The Koszul Formula

An elegant coordinate-free formula for the Levi-Civita connection, useful for abstract computations, is the *Koszul formula*:

$$2g(\nabla_X Y, Z) = X(g(Y, Z)) + Y(g(Z, X)) - Z(g(X, Y)) + g([X, Y], Z) - g([Y, Z], X) + g([Z, X], Y) \quad (\text{D.16})$$

This formula expresses the Levi-Civita connection entirely in terms of the metric g and Lie brackets $[X, Y]$ —neither of which requires a connection to be defined. The Koszul formula directly implies both torsion-freeness and metric compatibility, and provides a self-contained existence proof. It also makes the uniqueness manifest: given g , the right-hand side of (D.16) is completely determined for all X, Y, Z , so $\nabla_X Y$ is uniquely determined by the requirement that $g(\nabla_X Y, Z)$ equals the prescribed value for all Z (using non-degeneracy of g).

D.5 — Physical Significance of Each Condition

Torsion-freeness: On a torsion-free manifold, infinitesimally displaced parallelograms close up to second order in displacement. Consider moving from point p by εX then εY , versus moving by εY then εX —the two paths arrive at the same point to first order in ε . The torsion tensor $T^\rho_{\mu\nu}$ measures the failure of this closure. In standard GR, torsion-free means there is no intrinsic rotation of the reference frame during parallel transport. In Einstein-Cartan theory, torsion is sourced by the spin angular momentum density of matter. In ToE, the Levi-Civita connection of the Obidi Metric is torsion-free by definition: torsion effects on the statistical manifold Λ would correspond to a

non-trivial spin structure of the entropic field, which lies beyond the scope of the present Letter.

Metric compatibility: Means that lengths and angles are preserved under parallel transport along any curve. If V and W are parallel-transported along a curve γ , then $g(V, W)$ = constant along γ . This is the geometric prerequisite for a consistent notion of rigid measurement in curved space. Without metric compatibility, the "length" of a vector could depend on the path taken, making geometric measurement ambiguous and physics observer-dependent in an uncontrolled way.

D.6 — Connection to the Geodesic Equation

A curve $\gamma(\tau)$ on (M, g) is a *geodesic* if its tangent vector is parallel along itself: $\nabla_{\gamma'} \gamma' = 0$. In coordinates, with $x^\mu(\tau)$ the coordinate functions of γ :

$$d^2x^\mu/d\tau^2 + \Gamma^\mu_{\nu\rho} (dx^\nu/d\tau)(dx^\rho/d\tau) = 0 \quad (D.17)$$

The Christoffel symbols appearing here are precisely those of the Levi-Civita connection. The uniqueness theorem guarantees that (D.17) defines a *unique* family of geodesics for a given metric — the paths of freely falling observers in GR, and the paths of "least entropic resistance" in ToE.

Appendix E — Supplementing §5.5

How Fiber Integrals Produce Effective Fields — The Complete General Theory

E.1 — Setup: The Fiber Bundle Framework

Let $\pi: E \rightarrow B$ be a smooth fiber bundle with total space E , base manifold B (dimension n), and fiber F (dimension m). Let $\Omega^k(E)$ denote the space of smooth k -forms on E and $\Omega^j(B)$ the space of smooth j -forms on B .

The *fiber integral* (integration along the fiber) is a linear map:

$$\pi_*: \Omega^k(E) \rightarrow \Omega^{k-m}(B) \text{ for } k \geq m \quad (\text{E.1})$$

It "integrates out" the m fiber directions, reducing a k -form on E to a $(k-m)$ -form on B . The fiber integral is the mathematical operation of coarse-graining: it eliminates microscopic (fiber) degrees of freedom to produce effective macroscopic (base) objects.

E.2 — Construction of the Fiber Integral

Choose a local trivialization: on an open set $U \subset B$, we have $\pi^{-1}(U) \cong U \times F$. On this trivialization, let $x = (x^I, \dots, x^n)$ be local coordinates on B and $y = (y^J, \dots, y^m)$ be fiber coordinates. A general k -form $\omega \in \Omega^k(E)$ decomposes over this trivialization as:

$$\omega = \sum_{|I|+|J|=k} \omega_{IJ}(x, y) dx^I \wedge dy^J \quad (\text{E.2})$$

where $I = (i_1, \dots, i_{|I|})$ is an ordered multi-index of base directions, $J = (j_1, \dots, j_{|J|})$ is an ordered multi-index of fiber directions, and $dx^I = dx^{i_1} \wedge \dots \wedge dx^{i_{|I|}}$.

The fiber integral selects only the terms saturating all fiber directions ($|J| = m$):

$$(\pi_*\omega)_I(x) = \int_F \omega_{I,(1,\dots,m)}(x, y) dy^1 \wedge \dots \wedge dy^m \quad (\text{E.3})$$

for multi-indices I with $|I| = k - m$. Terms with $|J| < m$ do not contribute to the fiber integral (they do not have enough fiber directions to integrate). The result is a $(k-m)$ -form on B .

E.3 — The Projection Formula

Theorem E.1 (Projection Formula)

For any $\alpha \in \mathcal{Q}(B)$ and $\omega \in \Omega^k(E)$:

$$\pi_*(\pi^*\alpha \wedge \omega) = \alpha \wedge \pi_*(\omega)$$

A form pulled back from the base can be factored out of the fiber integral.

Proof.

In local coordinates, $\pi^*\alpha = \alpha_I(x) dx^I$ (the pullback introduces no dy components). Therefore $\pi^*\alpha \wedge \omega$ has the same fiber components as ω , multiplied by the function $\alpha_I(x)$ which depends only on x and can be factored out of the y -integration:

$$\pi_*(\pi^*\alpha \wedge \omega)_{IJ}(x) = \alpha_I(x) \int_F \omega_{J,(1,\dots,m)}(x,y) dy^1 \dots dy^m = (\alpha \wedge \pi_*\omega)_{IJ}(x)$$

(E.4)



E.4 — Stokes' Theorem for Fiber Integrals

The fiber integral intertwines with the exterior derivative via the identity:

$$d_B(\pi_*\omega) = \pi_*(d_E\omega) + \text{boundary terms}$$

(E.5)

If the fiber F is compact without boundary (e.g., $F = S^{m-1}$), or if all forms are compactly supported in the interior of F , the boundary terms vanish:

$$d_B \circ \pi_* = \pi_* \circ d_E \text{ (on compactly supported forms)} \quad (\text{E.6})$$

This is the fiber-integral version of Stokes' theorem. It converts integration-by-parts arguments on the total space E into clean algebraic identities on the base B . In ToE, this identity is used to convert the Boltzmann equation (a kinetic equation on phase space $E = M \times T^*M$) into the conservation equations (moment equations on the base M).

E.5 — How Moments Arise as Fiber Integrals: The Systematic Theory

Let the fiber be momentum space $F = \mathbb{R}^m$ (or more precisely, the mass shell), and let $f(x, p)$ be a smooth function on the total space $E = B \times \mathbb{R}^m$ — the one-particle distribution function. The k -th moment tensor of f is:

$$M^{\mu_1 \dots \mu_k}(x) = \int_{\mathbb{R}^m} p^{\mu_1} \cdot \dots \cdot p^{\mu_k} \cdot f(x, p) d\mu(p) \quad (\text{E.7})$$

where $d\mu(p)$ is the invariant measure on momentum space. This is the fiber integral of the k -form:

$$\omega^{(k)} = p^{\mu_1} \cdot \dots \cdot p^{\mu_k} \cdot f(x, p) \cdot d\mu(p) \quad (\text{E.8})$$

over the fiber $\mathbb{R}^m$. The result $M^{\mu_1 \dots \mu_k}(x)$ is a symmetric $(k, 0)$ tensor field on the base B . The effective field equations for these moment tensors are obtained by taking the fiber integral of the Boltzmann kinetic equation over momentum space.

E.6 — The Coarse-Graining Theorem

Theorem E.2 (Coarse-Graining via Fiber Integration)

Let $L(x, y, \partial_x \varphi, \partial_y \varphi)$ be a Lagrangian density on $E = B \times F$ for a field $\varphi(x, y)$. The effective Lagrangian on B obtained by integrating out the fiber is:

$$L_{\text{eff}}(x, \partial_x \varphi_{cl}) = \int_F L(x, y, \partial_x \varphi_{cl}, \partial_y \varphi_{\text{saddle}}(x, y)) dy^1 \dots dy^m$$

where $\varphi_{\text{saddle}}(x, y)$ is the saddle-point configuration in the fiber direction at fixed x , satisfying $\delta L / \delta \varphi_y = 0$.

This theorem is the mathematical foundation of Wilsonian effective field theory: integrating out high-energy (short-wavelength, fiber) degrees of freedom produces an effective Lagrangian for the low-energy (long-wavelength, base) degrees of freedom. The saddle-point approximation becomes exact in the classical limit.

ToE Application of Theorem E.2

In ToE: the base B is physical spacetime M ; the fiber F is the statistical manifold Λ_x of microscopic entropic configurations at each spacetime point x ; the total space is $E = M \times \Lambda$; the field φ is the entropy field $S(x, y)$ defined on E ; and the fiber integral produces the Obidi Action on M — the effective field theory of macroscopic entropic physics. The saddle-point condition $\delta L / \delta \varphi_y = 0$ is the *microscopic entropic equilibrium condition* which, in the near-equilibrium limit, reproduces the Einstein field equations.

E.7 — Application: Fiber Integral Derivation of $T^{\mu\nu}$

Beginning from the Boltzmann action on phase space $E = M \times T^*M$:

$$A_{Boltz} = \int_E f(x, p) [p^\mu \partial_\mu S - H(x, p)] dV_E \quad (\text{E.9})$$

where $H(x, p) = \sqrt{(g^{\mu\nu} p_\mu p_\nu + m^2 c^2)}$ is the relativistic Hamiltonian (mass-shell constraint). Integrating out the momentum fiber $T^*_x M$ at fixed x :

Step 1 — Saddle-point condition ($\delta A_{Boltz} / \delta f = 0$):

$$p^\mu \partial_\mu S = H(x, p) \text{ [the relativistic mass-shell condition]} \quad (\text{E.10})$$

Step 2 — Second-moment fiber integral:

$$T^{\mu\nu}(x) = \int_{T^*_x M} p^\mu p^\nu f(x, p) \cdot (\sqrt{-g} d^3 p / p^0) \quad (\text{E.11})$$

This is the stress-energy tensor — it is the output of the fiber integration of the second moment of f over momentum space. The derivation makes the fiber-integral origin of $T^{\mu\nu}$ fully explicit and manifestly Lorentz-covariant: the measure $\sqrt{-g} d^3 p / p^0$ is the invariant momentum-space measure on the mass shell.

Appendix F — Supplementing §6.5

Higher Moments and Radiation — The Complete Theory

F.1 — The Moment Hierarchy

For a relativistic distribution function $f(x, p)$ on phase space, with Lorentz-invariant measure $d\mu_p = \sqrt{-g} d^4p \cdot \delta(p^\mu p_\mu + m^2) \cdot \theta(p^0) / (2\pi)^3$, the complete moment hierarchy is:

$$N^\mu(x) = \int p^\mu f \cdot d\mu_p \text{ [four-current: rank-1]} \quad (\text{F.1})$$

$$T^{\mu\nu}(x) = \int p^\mu p^\nu f \cdot d\mu_p \text{ [energy-momentum tensor: rank-2]} \quad (\text{F.2})$$

$$Q^{\mu\nu\rho}(x) = \int p^\mu p^\nu p^\rho f \cdot d\mu_p \text{ [heat-flux tensor: rank-3]} \quad (\text{F.3})$$

$$R^{\mu\nu\rho\sigma}(x) = \int p^\mu p^\nu p^\rho p^\sigma f \cdot d\mu_p \text{ [viscosity tensor: rank-4]} \quad (\text{F.4})$$

All of these are symmetric tensors. The moment hierarchy does not close at any finite order: the equation governing the k -th moment involves the $(k+1)$ -th moment. Closure requires a truncation scheme (see §H.4–H.5 for the Chapman-Enskog approach).

F.2 — The Third Moment: Energy Flux and Heat Conduction

Let u^μ be the fluid four-velocity (normalized: $g_{\mu\nu}u^\mu u^\nu = -1$) and define the spatial projector $\Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$. The covariant decomposition of the third-rank moment tensor $Q^{\mu\nu\rho}$ with respect to u^μ gives:

$$Q^{\mu\nu\rho} = q^\mu u^\nu u^\rho + q^\nu u^\mu u^\rho + q^\rho u^\mu u^\nu + (\text{pressure-flow terms}) + (\text{isotropic terms}) \quad (\text{F.5})$$

where the *heat flux four-vector* is the projection:

$$q^\mu = -\Delta^{\mu\nu} u^\rho Q_{\nu\rho\sigma} u^\sigma \quad (\text{F.6})$$

In the near-equilibrium (Chapman-Enskog) expansion at first order in the Knudsen number, the heat flux satisfies Fourier's law in covariant form:

$$q^\mu = -\kappa \Delta^{\mu\nu} \nabla_\nu T \quad (\text{F.7})$$

where κ is the thermal conductivity and T is the local temperature. Equation (F.7) is not postulated — it is derived from the third moment of the Boltzmann distribution function at first order in the Knudsen number expansion. Thermal conductivity is therefore a property of the third moment of the momentum distribution.

F.3 — The Fourth Moment and Viscosity

The symmetric traceless part of the fourth-rank moment tensor $R^{\mu\nu\rho\sigma}$, projected onto the spatial directions, gives the *viscous stress tensor*. In the Chapman-Enskog expansion at first order in Knudsen number:

$$\Pi^{\mu\nu} = -\eta(\nabla^\mu u^\nu + \nabla^\nu u^\mu - (2/3)\Delta^{\mu\nu}\nabla_\rho u^\rho) - \zeta \Delta^{\mu\nu}\nabla_\rho u^\rho \quad (\text{F.8})$$

where η is the *shear viscosity* (resistance to shear flow) and ζ is the *bulk viscosity* (resistance to volume change). Both are derived from the fourth moment of the distribution function: η from the traceless symmetric part of $R^{\mu\nu\rho\sigma}$ and ζ from its trace. Viscosity is a fourth-order moment phenomenon.

F.4 — Radiation: The Specific Intensity and Its Moments

For a photon gas (massless particles, $p^\mu p_\mu = 0$), the momentum four-vector is parameterized by direction and frequency: $p^\mu = (h\nu/c)(1, n^i)$, where n^i is a unit

spatial direction vector and ν is the photon frequency. The *specific intensity* $I(x, n, \nu)$ is related to the photon distribution function by:

$$I(x, n, \nu) = (2h\nu^3/c^2) f(x, p(n, \nu)) \quad (\text{F.9})$$

The moments of the radiation field over direction and frequency are:

$$E_{rad}(x) = (1/c) \int_0^\infty \int_{S^2} I(x, n, \nu) d\Omega d\nu \text{ [radiation energy density]} \quad (\text{F.10})$$

$$\vec{F}_{rad}(x) = \int_0^\infty \int_{S^2} n^i I(x, n, \nu) d\Omega d\nu \text{ [radiation flux vector]} \quad (\text{F.11})$$

$$P_{rad}^{ij}(x) = (1/c) \int_0^\infty \int_{S^2} n^i n^j I(x, n, \nu) d\Omega d\nu \text{ [radiation pressure tensor]} \quad (\text{F.12})$$

The full stress-energy tensor of the radiation field, in block matrix form:

$$T_{rad}^{\mu\nu} = \begin{pmatrix} E_{rad} & \vec{F}_{rad}/c \\ \vec{F}_{rad}/c & P_{rad}^{ij} \end{pmatrix} \quad (\text{F.13})$$

F.5 — The Eddington Factor and Closure Relations

The *Eddington factor tensor* is the normalized second moment of the directional intensity distribution:

$$f^{ij}(x) = P_{rad}^{ij}(x) / E_{rad}(x) \quad (\text{F.14})$$

Its trace satisfies $\delta_{ij} f^{ij} = 1$ (since $\delta_{ij} P_{rad}^{ij} = (1/c) \int \delta_{ij} n^i n^j I d\Omega d\nu = E_{rad}$ using $|n|^2 = 1$). The eigenvalues of f^{ij} lie in $[0, 1]$ and sum to 1.

Two limiting cases:

- **Isotropic radiation** (thermal equilibrium, $I = I_0$): $f^{ij} = (1/3)\delta^{ij}$ — the Eddington limit.
- **Unidirectional beam** (free streaming in direction n_0): $f^{ij} = n_0^i n_0^j$ — maximum anisotropy.

The *variable Eddington factor closure* truncates the moment hierarchy by positing:

$$P_{rad}^{ij} = f^{ij} E_{rad} \quad (\text{F.15})$$

with f^{ij} determined by a local prescription. The Levermore-Pomraning closure and the M1 closure are standard choices. In ToE: the radiation Eddington factor is identified with the normalized second moment of the directional entropy distribution on the statistical manifold. The transition from isotropic ($f^{ij} = \delta^{ij}/3$) to anisotropic ($f^{ij} \neq \delta^{ij}/3$) corresponds to a transition from the equilibrium Fisher-Rao geometry to the non-equilibrium Amari-Čencov geometry of the entropic field.

F.6 — The Radiative Transfer Equation as a Moment Equation

The radiative transfer equation (Boltzmann equation for photons) is:

$$(1/c) \partial_t I + n^i \partial_i I = j_\nu - \kappa_\nu I \quad (\text{F.16})$$

where j_ν is the emissivity and κ_ν is the frequency-dependent absorption coefficient. This is the kinetic equation on the total space $E = M \times S^2$ (phase space with the fiber being the direction sphere). Taking fiber integrals over angle and frequency:

Zeroth angular moment ($\int d\Omega d\nu$ of (F.16)):

$$\partial_t E_{rad} + \nabla \cdot F_{rad} = c(\langle j \rangle - \bar{\kappa} E_{rad}) \text{ [radiation energy conservation]}$$

(F.17)

First angular moment ($\int n^i d\Omega dv$ of (F.16)):

$$(1/c^2)\partial_t \vec{F}_{rad} + \partial_j \vec{P}_{rad}^j = -(\bar{\kappa}/c)\vec{F}_{rad} \text{ [radiation flux evolution]}$$

(F.18)

These are the radiation hydrodynamics equations — effective field equations on the base M obtained by fiber integration of the radiative transfer equation over the momentum fiber S^2 . They provide the macroscopic description of radiation transport from the underlying kinetic (microscopic) description, in complete analogy with the derivation of the Navier-Stokes equations from the Boltzmann equation.

Appendix G — Supplementing §6.6

Why Physical Tensors Are Moment Objects — The Systematic Theory

G.1 — The Central Theorem

Theorem G.1 (Tensor-Moment Correspondence)

Every symmetric $(r, 0)$ tensor field $T^{\mu_1 \dots \mu_r}(x)$ arising as a conserved current in a local field theory can be expressed as the r -th moment of a phase-space distribution function $f(x, p)$:

$$T^{\mu_1 \dots \mu_r}(x) = \int_{T^*xM} p^{\mu_1} \cdot \dots \cdot p^{\mu_r} \cdot f(x, p) \cdot d\mu(p)$$

Proof sketch.

By the Schwartz nuclear theorem, any multilinear functional on test functions in the momentum variable can be represented as an integral kernel. For symmetric tensors built from bilinear expressions in conserved currents (energy, momentum), the integral kernel is precisely the distribution function $f(x, p)$. The symmetry of $p^{\mu_1} \dots p^{\mu_r}$ under index permutation guarantees the symmetry of T . The conservation condition $\partial_\mu T^{\mu\nu_2 \dots \nu_r} = 0$ is equivalent to the Liouville equation (collisionless Boltzmann equation) $p^\mu \partial_\mu f = 0$ for f , which follows from the chain rule applied to the moment integral: $\partial_\mu T^{\mu\nu_2 \dots \nu_r} = \int p^\mu p^{\nu_2} \dots p^{\nu_r} (\partial_\mu f) d\mu(p) = \int p^{\nu_2} \dots p^{\nu_r} (p^\mu \partial_\mu f) d\mu(p) = 0$.

**G.2 — Why the Metric Tensor Itself Is a Moment Object**

The spacetime metric $g_{\mu\nu}$ can be expressed as a zeroth moment of the statistical distribution over frame fields (vierbeins). Denoting the ensemble of microscopic entropic states by ω and the associated vierbein field by $e^a_\mu(x, \omega)$:

$$g_{\mu\nu}(x) = \int \eta_{ab} e^a_\mu(x, \omega) e^b_\nu(x, \omega) d\mu(\omega)$$

(G.1)

where $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ is the Minkowski metric and $d\mu(\omega)$ is the statistical measure over the entropic microstate space. This makes the spacetime metric a bilinear (hence zeroth-order, quadratic) moment of the frame distribution.

This is the precise content of the claim that "spacetime geometry is emergent from the statistical structure of the entropic field." In ToE, the Obidi Metric $\tilde{g}_{\mu\nu}$ is defined by an expression of the form (G.1), where the integral is the fiber integral over the statistical manifold Λ and the Obidi Transformation modifies the measure $d\mu(\omega)$.

G.3 — The Energy-Momentum Tensor as Second Moment in Fourier Space

For a scalar field φ on a flat spacetime with Lagrangian $L[\varphi, \partial_\mu \varphi]$, the canonical energy-momentum tensor derived from Noether's theorem (translation invariance) is:

$$T^{\mu\nu} = (\partial L / \partial(\partial_\mu \varphi)) \partial^\nu \varphi - g^{\mu\nu} L \quad (\text{G.2})$$

For a free scalar field with $L = -(1/2)(\partial_\mu \varphi \partial^\mu \varphi + m^2 \varphi^2)$, the Fourier transform $\tilde{\varphi}(k) = \int \varphi(x) e^{ik \cdot x} d^4x$ gives:

$$T^{\mu\nu}(x) = \int d^4k / (2\pi)^4 \cdot k^\mu k^\nu |\tilde{\varphi}(k)|^2 \quad (\text{G.3})$$

(after appropriate Fourier convolution). This makes explicit that $T^{\mu\nu}$ is the second moment of the spectral power distribution $|\tilde{\varphi}(k)|^2$ over momentum space: it is weighted by $k^\mu k^\nu$, exactly as in equation (E.7) with $f = |\tilde{\varphi}|^2$. The connection between the Noether tensor (G.2) and the kinetic moment (E.11) is thus explicit in Fourier space.

G.4 — The Riemann Tensor as a Fourth Moment

The Riemann curvature tensor $R_{\mu\nu\rho\sigma}$ can be expressed as a fourth-order covariance of connection fluctuations. Let $\Delta \Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\mu\nu} - \langle \Gamma^{\rho}_{\mu\nu} \rangle$ be the fluctuation of the Christoffel symbols around their ensemble average. Then:

$$R_{\mu\nu\rho\sigma} = \langle (\Delta \Gamma)_{\mu\rho} (\Delta \Gamma)_{\nu\sigma} \rangle - \langle (\Delta \Gamma)_{\mu\sigma} (\Delta \Gamma)_{\nu\rho} \rangle \quad (\text{G.4})$$

where $\langle \cdot \rangle$ denotes the average over the statistical ensemble of metrics. In the purely classical theory of GR, where there is a single deterministic metric, this expression requires interpretation: the ensemble average becomes an

average over the entropic microstate distribution. In ToE, the Riemann tensor of the Obidi Metric is exactly the fourth moment of the entropic fluctuation distribution — this is why the Einstein tensor $G_{\mu\nu}$ (built from the Riemann tensor) appears on the left-hand side of the ToE field equations as a fourth-order moment structure, while $T^{\mu\nu}_{ent}$ (the second moment) appears on the right-hand side.

G.5 — The Full Correspondence Table

Physical Tensor	Moment Order	Over Which Distribution
Particle number density n	0th	Momentum distribution $f(x, p)$
Four-current J^μ	1st	Momentum distribution $f(x, p)$
Stress-energy $T^{\mu\nu}$	2nd	Momentum distribution $f(x, p)$
Heat flux q^μ	3rd (contracted)	Momentum distribution $f(x, p)$
Viscous stress $\Pi^{\mu\nu}$	4th (contracted)	Momentum distribution $f(x, p)$
Radiation pressure P^j_{rad}	2nd	Directional intensity $I(x, n, \nu)$
Eddington factor f^j	2nd (normalized)	Directional intensity $I(x, n, \nu)$
Fisher-Rao metric G^F_{AB}	2nd	Score function distribution
Riemann tensor $R_{\mu\nu\rho\sigma}$	4th	Metric / connection fluctuations
Entropy S	0th (log-moment)	Microstate distribution $\{p_i\}$
Obidi Curvature Invariant (OCI)	0th	Binary choice distribution

Appendix H — Supplementing §6.7

The Kinetic Theory Connection — Distribution Functions, the Boltzmann Equation, and the Stress-Energy Tensor

H.1 — The One-Particle Distribution Function

In relativistic kinetic theory, the state of a many-particle system is described by the *one-particle distribution function* $f(x, p)$, defined so that:

$$f(x, p, t) d^3x d^3p = (\text{expected number of particles in } d^3x \text{ around } x, d^3p \text{ around } p, \text{ at time } t)$$
(H.1)

In the covariant (generally relativistic) formulation, f is defined on the *mass shell*:

$$\Sigma_m = \{(x, p) \in T^*M : g^{\mu\nu}(x) p_\mu p_\nu = -m^2 c^2, p^0 > 0\}$$
(H.2)

This is a 7-dimensional submanifold of the 8-dimensional cotangent bundle T^*M . The Lorentz-invariant measure on Σ_m is:

$$d\mu(p) = (\sqrt{-g} d^3p) / ((2\pi)^3 p^0) \text{ where } p^0 = \sqrt{(m^2 c^2 + g^{ij} p_i p_j)}$$
(H.3)

This measure is invariant under Lorentz transformations and diffeomorphisms of M . Its Lorentz invariance is the mathematical guarantee that kinetic theory is compatible with special and general relativity.

H.2 — The Boltzmann Equation

The time evolution of f is governed by the *relativistic Boltzmann transport equation*. On a curved spacetime manifold $(M, g_{\mu\nu})$:

$$p^\mu \partial_\mu f - \Gamma^\lambda_{\mu\nu} p^\mu p^\nu (\partial f / \partial p^\lambda) = C[f] \quad (\text{H.4})$$

The three terms have distinct physical meanings:

- **Free-streaming term:** $p^\mu \partial_\mu f$ — the rate of change of f along the phase-space flow of free (non-interacting) particles.
- **Gravitational (connection) term:** $-\Gamma^\lambda_{\mu\nu} p^\mu p^\nu (\partial f / \partial p^\lambda)$ — accounts for the curvature of spacetime accelerating particles (geodesic deviation in momentum space). The Christoffel symbols $\Gamma^\lambda_{\mu\nu}$ here are those of the Levi-Civita connection established in Appendix D.
- **Collision integral:** $C[f]$ — encodes particle-particle scattering processes.

For elastic binary collisions, the collision integral is:

$$C[f](x, p_1) = \int d\mu(p_2) d\mu(p_3) d\mu(p_4) |\mathcal{M}_{12 \rightarrow 34}|^2 \delta^4(p_1 + p_2 - p_3 - p_4) [f(p_3)f(p_4) - f(p_1)f(p_2)] \quad (\text{H.5})$$

where $|\mathcal{M}_{12 \rightarrow 34}|^2$ is the invariant scattering amplitude squared (from the quantum field theory of the interaction). The structure $f(p_3)f(p_4) - f(p_1)f(p_2)$ encodes the competition between gain (particles scattering into state p_1 from states p_3, p_4) and loss (particles in state p_1 scattering out to states p_3, p_4).

H.3 — The H-Theorem: Entropy Production from the Boltzmann Equation

The *Boltzmann H-function* is:

$$H[f] = \int f \ln(f) d\mu(p) dV \quad (\text{H.6})$$

Theorem H.1 (Boltzmann's H-Theorem)

For any solution $f(x, p, t)$ of the Boltzmann equation (H.4) with collision integral (H.5):

$$dH/dt \leq 0$$

with equality if and only if f is a local Maxwell-Boltzmann distribution (local thermodynamic equilibrium).

Proof sketch.

Differentiate H with respect to time using the Boltzmann equation. The free-streaming and connection terms contribute boundary terms that vanish for compactly supported f . The collision integral contributes:

$$dH/dt|_{\text{coll}} = \int C[f](1 + \ln f) d\mu(p) dV$$

(H.7)

By the symmetry of the collision integral under exchange of (p_1, p_2) and (p_3, p_4) (detailed balance, which follows from time-reversal invariance of the scattering amplitude $\mathcal{M}$) and the elementary inequality $(a-b)(\ln a - \ln b) \geq 0$ for all $a, b > 0$, one shows that (H.7) is non-positive. Equality holds if and only if $f(p_3)f(p_4) = f(p_1)f(p_2)$ for all collision events — i.e., if and only if $\ln f$ is a linear function of the conserved collision invariants (energy and momentum), which characterizes the Maxwell-Boltzmann distribution.



Since the entropy is $S = -k_B H$, the H-theorem is equivalent to $dS/dt \geq 0$ — the *second law of thermodynamics* is derived from the Boltzmann equation. The local entropy production rate is:

$$\sigma_s(x) = -k_B \int C[f] \ln(f) d\mu(p) \geq 0 \quad (\text{H.8})$$

and the local entropy balance equation is:

$$\partial_\mu s^\mu = \sigma_s \geq 0 \quad (\text{H.9})$$

where $s^\mu = -k_B \int p^\mu f \ln(f) d\mu(p)$ is the entropy four-current.

H.4 — Moment Equations from the Boltzmann Equation

Taking successive moments (fiber integrals over momentum) of the Boltzmann equation (H.4) yields the hydrodynamic conservation laws:

Zeroth moment — integrate (H.4) over $d\mu(p)$:

$$\nabla_\mu N^\mu = 0 \text{ [particle number conservation]} \quad (\text{H.10})$$

where $N^\mu(x) = \int p^\mu f d\mu(p)$ is the particle four-current. The right-hand side vanishes because the collision integral conserves particle number: $\int C[f] d\mu(p) = 0$.

First moment — integrate $p^\nu \times$ (H.4) over $d\mu(p)$:

$$\nabla_\mu T^{\mu\nu} = 0 \text{ [energy-momentum conservation]} \quad (\text{H.11})$$

where $T^{\mu\nu}(x) = \int p^\mu p^\nu f d\mu(p)$. The right-hand side vanishes because the collision integral conserves four-momentum: $\int p^\mu C[f] d\mu(p) = 0$.

Second moment — integrate $p^\nu p^\rho \times$ (H.4) over $d\mu(p)$:

$$\nabla_\mu Q^{\mu\nu\rho} = R^{\nu\rho} \text{ [heat flux equation]}$$

(H.12)

where $Q^{\mu\nu\rho}(x) = \int p^\mu p^\nu p^\rho f d\mu(p)$ is the third-rank moment and $R^{\nu\rho} = \int p^\nu p^\rho C[f] d\mu(p)$ is the source term from collisions (which does not vanish, since the collision integral does not conserve the second moment). This generates the unclosed hierarchy:

$$L_0 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow \dots$$

(H.13)

where L_k is the equation governing the k -th moment $T^{\mu_1 \dots \mu_k}$. Closure is achieved by truncation: the 14-moment Grad approximation, or systematically by the Chapman-Enskog expansion.

H.5 — The Chapman-Enskog Expansion

The Chapman-Enskog expansion expands the distribution function in powers of the Knudsen number $Kn = \ell/L$ (ratio of mean free path to macroscopic length scale):

$$f = f^{(0)} + Kn \cdot f^{(1)} + Kn^2 \cdot f^{(2)} + \dots$$

(H.14)

Zeroth order — $f^{(0)}$ is the local relativistic Maxwell-Jüttner distribution (local thermodynamic equilibrium):

$$f^{(0)}(x, p) = \exp[-\beta(x)(p^\mu u_\mu(x) + \mu(x))]$$

(H.15)

where $\beta(x) = 1/(k_B T(x))$ is the inverse local temperature, $u^\mu(x)$ is the fluid four-velocity, and $\mu(x)$ is the local chemical potential. At this order, the stress-energy tensor is that of a perfect fluid:

$$T^{\mu\nu}_{(0)} = (\varepsilon + p)u^\mu u^\nu + p g^{\mu\nu} \text{ [perfect fluid]}$$

(H.16)

where ε is the energy density and p is the thermodynamic pressure.

First order in Kn — $f^{(1)}$ encodes the deviation from local equilibrium due to spatial gradients. Substituting into the Boltzmann equation and solving the linearized collision operator gives the Navier-Stokes stress-energy tensor:

$$T^{\mu\nu}_{(1)} = -\eta(\nabla^\mu u^\nu + \nabla^\nu u^\mu + (2/3)g^{\mu\nu}\nabla_\rho u^\rho) - \xi g^{\mu\nu}\nabla_\rho u^\rho \text{ [viscous corrections]}$$

(H.17)

where η is the shear viscosity and ξ is the bulk viscosity. These are *derived* from the first-order correction $f^{(1)}$ by computing the second moment. The coefficients η and ξ are expressed as integrals of the collision kernel — they are properties of the microscopic scattering process, encoded in the distribution f . This derivation shows that viscosity is a first-order Knudsen-number correction to the momentum distribution; it is not a primitive fluid property but an emergent one.

H.6 — ToE Generalization: The Entropic Boltzmann Equation

In ToE, the distribution function $f(x, p)$ is replaced by the *entropic distribution function* $f_{ent}(x, q)$, where $q_\mu = \partial_\mu S$ is the *entropic momentum* — the gradient of the entropy field, a covector on the statistical manifold. The *Entropic Boltzmann Equation* is:

$$q^\mu \partial_\mu f_{ent} - \Gamma^{\lambda (F)}_{\mu\nu} q^\mu q^\nu (\partial f_{ent} / \partial q^\lambda) = C_{ent}[f_{ent}]$$

(H.18)

where $\Gamma^{\lambda (F)}_{\mu\nu}$ are the Christoffel symbols of the Fisher-Rao metric $G^F_{\mu\nu}$ on the statistical manifold (derived from the Levi-Civita connection of G^F via the formula of Appendix D), and $C_{ent}[f_{ent}]$ is the entropic collision integral encoding irreversible entropy production.

The moment equations of (H.18) give:

$$\nabla_\mu N^\mu_{ent} = 0 \text{ [entropic number conservation]} \quad (\text{H.19})$$

$$\nabla_\mu T^{\mu\nu}_{ent} = 0 \text{ [entropic energy-momentum conservation]} \quad (\text{H.20})$$

In the near-equilibrium limit, the Entropic Stress-Energy Tensor approaches:

$$T^{\mu\nu}_{ent} \rightarrow (c^4 / 8\pi G) G^{\mu\nu} \quad (\text{H.21})$$

where $G^{\mu\nu} = R^{\mu\nu} - (1/2)g^{\mu\nu}R$ is the Einstein tensor. Equation (H.20) with (H.21) substituted gives:

$$G^{\mu\nu} = (8\pi G / c^4) T^{\mu\nu}_{ent} \quad (\text{H.22})$$

which are the Einstein field equations. Thus: *the Einstein field equations are the first-moment equations of the Entropic Boltzmann Equation in the near-equilibrium limit.*

H.7 — Summary: The Path from Kinetic Theory to the Einstein Field Equations via ToE

Derivation Chain: Entropic Kinetic Theory → Einstein Field Equations

Step 1. Specify the entropic distribution function $f_{ent}(x, q)$ on the statistical manifold Λ equipped with the Fisher-Rao metric G^F_{AB} .

Step 2. Take the zeroth moment of the Entropic Boltzmann Equation →

entropic particle number conservation (H.19).

Step 3. Take the first moment of the Entropic Boltzmann Equation $\rightarrow$ entropic energy-momentum conservation: $\nabla_\mu T^{\mu\nu}_{ent} = 0$ (H.20).

Step 4. Apply the Obidi Transformation $\mathcal{T}_{Obidi}$ to obtain the Obidi Metric $\tilde{g}_{\mu\nu}$ from the Fisher-Rao metric G^F_{AB} (via the pullback formula of §B.5).

Step 5. Compute the Einstein tensor $G_{\mu\nu}[\tilde{g}]$ from the Obidi Metric via the Riemann tensor and Levi-Civita connection of Appendix D.

Step 6. In the near-equilibrium limit, $T^{\mu\nu}_{ent} \rightarrow (c^4/8\pi G)G^{\mu\nu}$, recovering the Einstein field equations (H.22).

This completes the derivation of the Einstein field equations as moment equations of the Entropic Boltzmann Equation. The physical interpretation: the Einstein field equations describe how spacetime curvature responds to the distribution of energy and momentum — they are the statement that the first moment of the entropic kinetic distribution on the statistical manifold, in near-equilibrium, equals (up to constants) the geometric (Einstein) tensor of the emergent spacetime metric. Gravity, in ToE, is the first-moment response of the entropic kinetic theory to macroscopic matter distributions.

Coverage Map: TOC Sections vs. Appendix Coverage

The following table summarizes the coverage status of each supplemented section, the gap addressed by the corresponding appendix, and the key mathematical content supplied:

TOC Section	TOC Title	Main Text Coverage	Appendix	Key Content Supplied
§2.6	Mass, Pressure, Radiation, and Stress Not Alien to Information	Absent from main text	Appendix A	Moment-theoretic derivations of all four quantities; Fundamental Principle; unified table
§3.3	Smooth Maps, Diffeomorphisms, and Pullbacks	Pullback mentioned only in passing; diffeomorphism invariance stated without proof	Appendix B	Full theory of smooth maps, Jacobian, pushforward, immersions/embeddings, diffeomorphisms, pullback component formula, musical isomorphisms, Lie derivative, Killing equation
§4.1	Scalars, Vectors, Covectors — Unified View	Partial: classification given, algebraic structure absent	Appendix C	Tensor product algebra, universal property, complete classification table, musical isomorphisms, contraction, entropy field hierarchy (Levels 0–2)

TOC Section	TOC Title	Main Text Coverage	Appendix	Key Content Supplied
§4.9	Levi-Civita Connection — Uniqueness	Christoffel formula stated; proof of uniqueness absent	Appendix D	Complete proof of existence and uniqueness (Fundamental Theorem); Koszul formula; physical significance of torsion-free and metric-compatible conditions; geodesic equation
§5.5	How Fiber Integrals Produce Effective Fields	Examples given; general theory absent	Appendix E	Fiber bundle setup, fiber integral construction, projection formula (proved), Stokes' theorem for fiber integrals, coarse-graining theorem, fiber integral derivation of $T^{\mu\nu}$
§6.5	Higher Moments and Radiation	Third moment only; radiation absent	Appendix F	Complete 4-level moment hierarchy; heat flux decomposition; viscosity

TOC Section	TOC Title	Main Text Coverage	Appendix	Key Content Supplied
				tensor; specific intensity; radiation energy/flux/ pressure moments; Eddington factor; closure relations; radiation hydrodynam- ics equations
§6.6	Why Physical Tensors Are Moment Objects	Partially covered for $T^{\mu\nu}$ only	Appendix G	Tensor- Moment Correspon- dence Theorem (proved); metric as zeroth moment; $T^{\mu\nu}$ as second Fourier moment; Riemann tensor as fourth moment; full correspon- dence table (11 tensors)
§6.7	Kinetic Theory Connection	Formula for $T^{\mu\nu}$ given; Boltzmann equation absent	Appendix H	Full relativistic Boltzmann equation; H- theorem (proved); entropy production; moment hierarchy derivation (3

TOC Section	TOC Title	Main Text Coverage	Appendix	Key Content Supplied
				orders); Chapman-Enskog expansion; Entropic Boltzmann Equation; derivation of Einstein field equations as moment equations

Total coverage: Eight appendix sections (A–H) supply complete mathematical content for eight sections of the Letter IV Part I Table of Contents. All proofs are complete. All physical interpretations are connected to the Theory of Entropicity. The document is self-contained and may be read independently of the main text.

End of Mathematical Appendix — ToE Letter IV, Part I

Author: John Onimisi Obidi | Research Lab: The Aether | Date: June 6, 2026

Canonical Archive: <https://entropicity.github.io/Theory-of-Entropicity-ToE/>

Theory of Entropicity (ToE) Living Review Letters Series — Letter IV, Part I — Mathematical Appendix